

CS 570: Algorithms for Curves and Surfaces

Course Code: CS 570

Course Name: Algorithms for Curves and Surfaces

L-T-P-C: 2-0-2-3

Prerequisites: IC 111 Linear Algebra or equivalent course, IC 250 Data Structure and Algorithms or equivalent course.

Intended for: B.Tech. (CSE, DSE, EE and ME) II/III/IV year/ M.Tech/M.Tech(R)/Ph.D (any branch)

Mutual Exclusion: None

Approval: 61st BoA

Preamble

This course is intended to introduce the algorithms and techniques found in modern CAD kernels. The primary requirement of such algorithms is a high degree of numerical precision. The main application area of this technology is in engineering design. The course will include assignments about using popular CAD software suites as well as programming assignments.

Course Modules

- **Introduction:** Motivation; Use of CAD software in different domains; Example use-cases; Different geometric representations. (2 Hours)
- **Differential geometry of curves:** Parametric curves, tangent-spaces, curvature, Frenet-Serret equations, winding number of a planar curve. (4 Hours)
- **Differential geometry of surfaces:** Parametric surfaces, tangent-spaces, normals, the first fundamental form, lengths of curves on surfaces, the second fundamental form, the Gauss and Weingarten maps, Gaussian and mean curvatures, Principal curvatures. (6 Hours)
- **Bezier curves and surfaces:** Definition and properties of Bezier curves, Bernstein basis, De Casteljau's algorithm, definition and properties of Bezier surfaces, Subdivision algorithm. (6 Hours)
- **B-splines:** Definition and construction. de Boor's algorithm for evaluation. (4 Hours)
- **Boolean operations:** Boundary representation, Euler's formula, curve-curve intersection, curve-surface intersection, surface-surface intersection, projection of a curve on a surface. (6 Hours)

Laboratory/Practical/Tutorial Modules

- **Lab 1:** Part design using a CAD software. Emphasis on operations such as 2D sketching, extruding, blending and booleans.
- **Lab 2-4:** Plot parametric curves and their tangents, normals, binormals, osculating circles at sampled points. This and all subsequent labs are to be done in C++ using OpenCascade geometric kernel.
- **Lab 5-7:** Plot parametric surfaces and their tangent planes, normals, directions of principal curvatures. Computing Gaussian and mean curvatures. Computing lengths of curves on surfaces.
- **Lab 8-10:** Write code to implement evaluation of Bezier curves and surfaces using De Casteljau's algorithm. Verify various properties Bezier curves and surfaces.

- **Lab 11-12:** Write code to compute intersection of a pair of planar parametric curves and for a pair of parametric surfaces.
- **Lab 13-14:** Write code to traverse the different topological entities in the Brep of a solid. Write code to project a space curve onto a surface.

Text Books

1. Gerald Farin, *Curves and Surfaces for CAGD: A Practical Guide*, 5th edition, Morgan Kaufmann Publishers, USA, 2002
2. Andrew Pressley, *Elementary Differential Geometry*, 2nd Edition, Springer, USA, 2010

References

1. Andrew Pressley, *Elementary Differential Geometry*, 2nd Edition, Dover Publications, USA, 2016